Contents

Forced Chirality: The Derived Fermionic Dynamics of $(\mathbb{S}, \tau)$, the Domain-Wall Sector, and the Orientation Residue

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The Phase 3 report. Companion to the unified paper (Rev. 1.2 + fourth-witness corollary), the Phase 3 plan (frozen, §§0–7), and the Foundation/Keystone chain. Every numerical claim below was computed in two independent implementations; every scalar instrument carries its defining formula (plan §7, P6); pre-registrations preceded every computation stage, and the deviations ledger is in the development notes.

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1. Charter

The question. Does this framework's fermionic sector admit a state whose polarization selects a chiral half of the matter module — by a *derived* dynamics, with a determinate relationship to the gauge action?

Scope, fixed in advance. "Chirality" throughout means the state-selected structure of the internal module; its identification with spacetime chirality (γ^5) is conditional on the framework's eventual spacetime account (the fork's Reading B, held open). The arena is quantum mechanics on the derived scale line — not field theory in spacetime. The e_R boundary binds every outcome: no half of this matter content contains a full generation.

2. The map (Stage 1)

[THM, two-implemented] The real commutants of the gauge chain on the 16-dimensional matter module have dimensions **24 / 16 / 4** (color / color+Y

/ full G_{SM}). Exactly **four** G_{SM}-compatible complex structures exist (two up to conjugation), and **every one selects a chiral** $Q \oplus L$ — one generation's left-handed doublet sector, $(\mathbf{3}, \mathbf{2}) \oplus (\mathbf{1}, \mathbf{2})$ with lepton/quark hypercharge ratio -3.000 — in four orientation variants splitting $2+2$ into an SM-shaped class (ratio -3) and a mixed class ($+3$).

[THM] General classification: on an isotypic block (complex-type irreducible W, multiplicity m), compatible complex structures are classified by **signature** (p, q), $p + q = m$; selected content is $p \cdot W \oplus q \cdot \bar{W}$; **chirality** $\iff$ **definite signature** $\iff$ **centrality in the block commutant**. The continuous strata are vectorlike or mixed; full gauge compatibility annihilates them, leaving only centers — Stage 1a as corollary.

Census discipline (adopted as a standing rule). Each compatibility class is censused only in the charges it preserves; compression spectra are diagnostics, never weights.

3. The menu and the graded legality rule (Stage 1c)

[FACT, two-implemented] Real (CAR-side) dynamics is orthogonal; legal generator terms obey **total-parity skew**: (odd line)⊗(symmetric internal) or (even line)⊗(antisymmetric internal). The internal menus: 16 operators at color+Y in eight uniform cells of two (parity $\times$ Ω-grading $\times$ block), 24 at color-only (6/2/4/4 on the lepton block); anchors $\Omega \in$(sym, even), central J's $\in$(antisym, even). $\Omega^\top = (-1)^k \Omega$ across representations with k antisymmetric generators; the physical case $k = 8$ gives Ω symmetric.

4. The keystone: the flow extends, and the odd half fermionizes (Stage 2a)

[THM, two-implemented] The derived flow extends to the odd half of $\mathbb{S}$ through a consistent graded contraction: with even grading $(1, X, P, W) = (0, 1, 2, 3)$, the odd weights are forced to the **mirrored pattern** b_0, $b_0{+}1$, $b_0{+}2$, $b_0{+}3$, admissible **iff** $b_0 \geq 0$ (27 deficits below, zero at or above). The extension is τ-even by construction.

[FACT, two-implemented] At the distinguished point $b_0 = 0$: all seven odd diagonal squares die (deficit values $2, 2, 2, 4, 4, 4, 6$); exactly the cross products survive ($X \times X \to P$, $X \times P \to W$) — a **Grassmann-type induced structure** (square-zero generators, surviving cross-brackets). Stated at the careful weight: the induced algebra is Grassmann-type; "CAR-ready" is a quantization-level claim earned in §8's construction, not by this sweep.

[THM, the ladder] All 19 surviving even-on-odd products satisfy $\text{level}(\text{out}) = \deg(\text{even}) + \text{level}(\text{in})$: the contracted even algebra acts as a four-rung **creation ladder** (X raises by 1, P by 2, W by 3, top annihilated in the contraction limit).

5. The expansion: what the derived dynamics knows (registered instruments)

Instruments (P6): *menu-fraction*(A) $:=$ $\|\Pi_{\mathcal{C}}(A)\|_F^2/\|A\|_F^2$ (Frobenius projection onto the commutant span); *commutator-profile*$(A, \mathfrak{g})$ $:=$ $\max_g \|[A, g]\|$.

[FACT, two-implemented, exact] The dilation's internal part A_{dil} has menu-fraction $15/16$ — breaking remainder exactly $\frac{1}{2}S_{PX}$, $1/16$ in norm. The squeeze's internal part is $A_{\text{sq}} = -S_{PX}$: **menu-fraction** 0 — the KMS-relevant flow's internal content is *purely* the P/X-flavor distinction, wholly outside the gauge-expressible span, vanishing identically on the lepton block. Its centralizer in color is **exactly the 3-dimensional** $SO(3)$ **remnant**.

[FACT] The legality tension. A_{sq} is symmetric — squeezes preserve the symplectic form, not the metric — so Id $\otimes\, A_{\text{sq}}$ is not a legal CAR generator: the bosonic/fermionic asymmetry (σ-preserving vs g-preserving time) is a structural fact of this framework, surfaced by its own menu. The squeeze enters fermionic dynamics only through parity-legal partners.

6. Family one and the trade-off theorem

[FACT, proven and verified] The naive candidate $\partial_u \otimes \mathrm{Id} + m(u) \otimes K$ ($K = \Omega J_0$; $K^2 = -\mathrm{Id}$, gauge-Cartan-commuting, grading-odd against $T_\pm$ exactly) is a **removable pure twist** — no gap, no wall modes; its content is an infrared winding of the weak plane. **[FACT]** $\{A_{\text{sq}}, K\} = 0$ globally (a Dirac pairing between the framework's two derived internal operators), and $[\,\mathrm{Id}{+}\mu A_{\text{sq}}, K\,]$ vanishes **exactly on the lepton block**: in the composite $\partial_u \otimes (\mathrm{Id} + \mu A_{\text{sq}}) + m(u) \otimes K$, *leptons wind, only quarks can bind*.

[THM, numerically certified] Wall localization occurs **iff** $\mu > 1$ (counter-propagation threshold) — certified by gap-scaling in system size and exponential mode localization. **[THM, trade-off]** At $\mu > 1$ the surviving symmetry is exactly $SO(3)_{\text{remnant}} \times \Omega$; the six wall modes form $(\mathbf{3}, \Omega{=}{+}) \oplus (\mathbf{3}, \Omega{=}{-})$, forced **real** by $SO(3)$'s representation theory. *In this family, localization and gauge preservation are mutually exclusive, and the wall cannot be chiral.*

7. The gate theorem

[THM, two-implemented, maximal] Ω-odd $\cap$ T_3-commuting $= 0$ identically (singular values exactly 1): because T_3's eigenspaces *are* the chirality columns ($T_3 = \lambda J$), **no operator can mix the chirality columns while preserving weak isospin**. **[INTERPRETATION]** Any mass binding the two chiralities necessarily breaks the weak Cartan — the Standard Model's own structural reason that fermion masses require electroweak breaking, re-derived here from the fork's identity in two lines. Tagged at interpretation weight; the algebra above it is exact.

8. Family two: the wall is chiral

[DEF/FACT] K' is drawn from the four-dimensional **(antisym, Ω-odd) gauge-covariant cell** (two elements per block — per-sector mass freedom, *leptons participate*). $\{\Omega, K'\} = 0$ gives the Dirac pairing with the legal velocity-splitter $\partial_u \otimes (\mathrm{Id} + \nu\Omega)$; $[K', T_3] \neq 0$ is forced (§7). Threshold $\nu > 1$; **eight** wall modes (2 lepton + 6 quark), internal amplitude ratio $(\nu{+}1)/(\nu{-}1)$ and localization rate $\kappa^2 = m_0^2/((\nu{-}1)(\nu{+}1))$ — both derived by hand independently on each side (one route via $\lambda^2 = -m_0^2/(v_1 v_2)$, the same formula) and verified numerically at $\nu = 1.5$ (5:1) and $\nu = 2$ (3:1, matching $(\nu{-}1)/(\nu{+}1) = 0.3333$ to four decimals); the per-line reduction is the J_0-complexified two-component Jackiw–Rebbi problem, with the mode-count halving being normalizability itself.

[THM, the invariant complex structure] $J_0 = -(Y/y)_{\mathrm{lep}} \oplus +(Y/y)_{\mathrm{quark}}$ exactly — the gauge-compatible complex structure *is* blockwise-normalized hypercharge — and J_0 is **central in the entire gauge commutant**, hence $[J_0, \mathfrak{A}] = 0$: the wall sector inherits J_0, and its census is well-posed.

[THM, forced chirality — Schur route] Color commutes with every operator defining the construction (Ω, K', J_0), so the zero-mode projection is color-equivariant and, by Schur's lemma, cannot partially dismantle the irreducible complex **3**; the wall's quark sector is a whole **3** (or $\bar{\mathbf{3}}$), and $\mathbf{3} \not\cong \bar{\mathbf{3}}$. **The wall content $\{\mathbf{1}_{y_\ell}\} \oplus \{\mathbf{3}_{y_q}\}$ is necessarily not self-conjugate — chiral, under the unbroken $SU(3) \times U(1)_Y$, independent of all spectral details beyond the threshold theorem's own existence claim.** The charter's criterion (ii) is met by a construction whose every ingredient is derived or menu-classified.

9. The orientation residue

[FACT] All four orientation structures $(\pm, \pm)$ commute with $\mathfrak{A}$ — the dynamics contains no J_0 and provably cannot prefer among them — so the zero-energy wall sector is orientation-degenerate: **whether the wall's chiral class is SM-shaped (ratio -3) or mixed ($+3$) is the one binary datum this phase's dynamics does not fix.** This is the polarization doctrine's own endpoint, not a late defect: the state must choose the orientation, and $\chi_{>0}$ is silent exactly at zero. Named successor question, candidate selectors listed (the $\beta \to \infty$ approach to the zero sector; the Foundation's settling structure), untried.

10. Verdict and honesty ledger

Verdict: outcome O2, as pre-priced in the frozen plan — selection succeeds; the gauge symmetry breaks spontaneously to $SU(3) \times U(1)_Y$; the wall-localized content is necessarily chiral; the breaking pattern and the mass/weak-isospin gate theorem are of independent interest.

Ledger, drafted before any softening could occur: the composite was selected by stated principles (gauge covariance, Dirac pairing, the family-one no-go), not a uniqueness theorem; ν and b_0 are free knobs with derived thresholds but underived values; the distinguished point $b_0 = 0$ is argued, not proven unique; the arena is the scale line — internal chirality, Reading B standing; the orientation residue is open; and the e_R/generation boundary is untouched by everything here. Open items inherited and new: the successor orientation question; the CAR quantization built explicitly on §4's Grassmann structure; the relation of the wall theory to the bosonic KMS chain's thermal structure; the fork's final adjudication.

Created in collaboration with Claude (Anthropic). All conclusions and any errors remain the author's responsibility.